

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

SE 19-206-0406 OCCUPATIONAL HEALTH AND FIRST AID

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain the concept and spectrum of health -functional units and activities of occupational health service.

CO2: Interpret physical chemical and biological hazards in the work environment and its control measures.

CO3: Recognize the principles of first aid.

CO4: Summarize the anatomy and functions of different human systems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
1.	(a) How do occupational health services contribute to worker well-being and productivity?	3	3	L2	1	1,6
	(b) Explain the levels of prevention strategies for occupational diseases.	3	3	L2	1	1
	(c) Describe heat exhaustion and its symptoms.	3	3	L1	2	1
	(d) What are the advantages of good illumination in a workplace?	3	3	L1	2	1
	(e) What is the sequence of actions on arrival at an emergency situation?	3	3	L2	3	1
	(f) What are tendons, ligaments and cartilage?	3	3	L1	3	1
	(g) What is the classification of wounds based on their depth?	3	3	L1	4	1
	(h) How do you differentiate sprain and fracture?	3	3	L2	4	1

PART B

(4 × 12 = 48)

II.	(a) Select three notifiable occupational diseases and explain their causes, symptoms and preventive measures.	6	6	L2	1	1
	(b) Describe the equipments used for radiation detection and measurement.	6	6	L1	1	1
OR						
III.	(a) Explain the importance of threshold limit values and discuss how they are calculated.	6	6	L2	1	1
	(b) Describe the noise effects and control methods.	6	6	L1	1	1
IV.	Describe about the vibrational health effects on human body.	12	12	L2	2	1
OR						
V.	Explain the following:					
	(i) Local exhaust ventilation.	4	4	L1	2	1
	(ii) WBGT Index.	4	4	L1	2	1
	(iii) Standards of illumination in workplace.	4	4	L1	2	1

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		Marks	BL	CO	PO
VI.	(a) What are the causes and symptoms of seizure?	6	L1	3	1
	(b) Explain the causes and levels of unconsciousness and how should they be managed.	6	L1	3	
OR					
VII.	How does the aim, objectives and principles guide the role of a first aider?	12	L1	3	1
VIII.	(a) Explain the causes, signs and symptoms of shock and how would you manage it as a first aider.	6	L1	4	1
	(b) Describe the steps involved in performing CPR on an adult.	6	L2	4	1
OR					
IX.	(a) What are different types of burns? How would you manage them as first aider?	6	L2	4	1
	(b) Explain the rule of nine to assess the severity of burns.	6	L1	4	1

Bloom's Taxonomy Levels

L1 - 60%, L2 - 40%.
